Decent Coaching Classes

X SB

Mind Your Math VIII — Answer Key & Marking Scheme

Subject : Mathematics Part II **Max. Marks : 40**

Q. 1 (A) Four alternative answers are given for every sub-question. Select the correct alternative. (4)

(i) The ratio of the areas of two similar triangles is 16 : 25. The ratio of their corresponding sides is

(A) 16 : 25 (B) 4 : 5 (C) 5 : 4 (D) 8 : 5

Ans: (B) 4 : 5 [Ratio of sides = √(16/25) = 4/5, i.e. 4 : 5.] (1)

(ii) The legs of a right-angled triangle are 9 cm and 12 cm. Its hypotenuse is

(A) 21 cm (B) 18 cm (C) 15 cm (D) 13 cm

Ans: (C) 15 cm [$\sqrt{(9^2 + 12^2)} = \sqrt{225} = 15$ cm.] (1)

(iii) The radius of a circle is 5 cm. The length of the tangent from a point 13 cm from the centre is

(A) 12 cm (B) 8 cm (C) 18 cm (D) 14 cm

Ans: (A) 12 cm [Tangent² = $13^2 - 5^2 = 144 \Rightarrow$ tangent = 12 cm.] (1)

(iv) The distance of the point (3, 4) from the origin is

(A) 7 (B) 1 (C) 12 (D) 5

Ans: (D) 5 [$\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.] (1)

Q. 1 (B) Solve the following sub-questions. (4)

(i) If $\sin\theta = 3/5$, find $\cos\theta$.

$\cos\theta = \sqrt{(1 - \sin^2\theta)} = \sqrt{(1 - 9/25)} = 4/5$ (1)

(ii) Find the slant height of a cone whose radius is 3 cm and height is 4 cm.

$l = \sqrt{(r^2 + h^2)} = \sqrt{(9 + 16)} = 5$ cm (1)

(iii) In a 30°–60°–90° triangle, the hypotenuse is 10 cm. Find the length of the side opposite to the 30° angle.

Side opposite 30° = ½ × hypotenuse = 5 cm (1)

(iv) The measure of an inscribed angle in a circle is 40°. Find the measure of its intercepted arc.

Arc = 2 × inscribed angle = 80° (1)

Q. 2 (A) Complete any two activities and rewrite them. (4)

(i) In ΔABC, points D and E lie on sides AB and AC respectively and DE ∥ BC. If AD = 4, DB = 6 and AE = 6, find EC.

In ΔABC, DE ∥ BC ∴ AD/DB = AE/EC (Basic proportionality theorem)

∴ 4/6 = 6/EC ∴ EC = 9

(each blank-set ½ mark; total 2)

(ii) In ΔPQR, ∠Q = 90°, PQ = 6 and QR = 8. Find PR.

$PR^2 = PQ^2 + QR^2$ (Pythagoras theorem)

∴ $PR^2 = 36 + 64 = 100$ ∴ PR = 10

(iii) Find the coordinates of the midpoint M of the segment joining A(2, 3) and B(8, 9).

$x = (x_1 + x_2)/2 = (2 + 8)/2 = 5$

$y = (y_1 + y_2)/2 = (3 + 9)/2 = 6$

∴ M(5 , 6)

Marking: each correct activity = 2 marks (any two attempted).

Q. 2 (B) Solve any two sub-questions. (4)

(i) Find the value of $\sin 60° \times \cos 30° + \tan 45°$.

$= (\sqrt{3}/2) \times (\sqrt{3}/2) + 1$ (1)

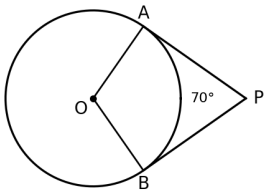
$= 3/4 + 1 = 7/4$ (1)

(ii) The radius of a cylinder is 7 cm and its height is 10 cm. Find its curved surface area. ($\pi = 22/7$)

Curved surface area $= 2\pi rh = 2 \times (22/7) \times 7 \times 10$ (1)

$= 440\ cm^2$ (1)

(iii) In the figure, PA and PB are tangents to the circle with centre O from the external point P. If $\angle APB = 70°$, find $\angle AOB$.



OA ⊥ PA and OB ⊥ PB (tangent–radius theorem), so $\angle OAP = \angle OBP = 90°$ (1)

In quadrilateral AOBP: $\angle AOB = 360° - (90° + 90° + 70°) = 110°$ (1)

Q. 3 (A) Complete any one activity and rewrite it. (3)

(i) $\Delta ABC \sim \Delta PQR$, $A(\Delta ABC) = 36$ sq. unit, $A(\Delta PQR) = 64$ sq. unit and $AB = 9$ units. Find PQ.

$A(\Delta ABC)/A(\Delta PQR) = AB^2/PQ^2$ (Theorem on areas of similar triangles)

$\therefore\ 36/64 = 81/PQ^2 \quad \therefore\ PQ^2 = (81 \times 64)/36 = 144 \quad \therefore\ PQ = 12$ units

(ii) Chords AB and CD of a circle intersect at point P inside the circle. If AP = 4, PB = 6 and CP = 3, find PD and CD.

$AP \times PB = CP \times PD$ (Theorem of internal division of chords)

$\therefore\ 4 \times 6 = 3 \times PD \quad \therefore\ PD = 8$

$CD = CP + PD = 3 + 8 = 11$

Marking: either activity = 3 marks (any one attempted).

Q. 3 (B) Solve any two sub-questions. (6)

(i) The diagonals of a rhombus are 16 cm and 12 cm. Find the side of the rhombus and its perimeter.

Diagonals of a rhombus bisect each other at right angles, so half-diagonals are 8 cm and 6 cm. (1)

$Side^2 = 8^2 + 6^2 = 64 + 36 = 100 \Rightarrow$ side = 10 cm (1)

Perimeter $= 4 \times 10 = 40$ cm (1)

(ii) Find the coordinates of the point P which divides the segment joining A(1, −2) and B(4, 7) internally in the ratio 1 : 2.

Section formula: $x = (m_1x_2 + m_2x_1)/(m_1 + m_2)$, $y = (m_1y_2 + m_2y_1)/(m_1 + m_2)$; $m_1 : m_2 = 1 : 2$ (1)

$x = (1 \times 4 + 2 \times 1)/3 = 6/3 = 2$ (1)

$y = (1 \times 7 + 2 \times (-2))/3 = 3/3 = 1$ (½)

Ans: P(2, 1) (½)

(iii) Prove that: $(1 + \tan^2\theta) \times \cos^2\theta = 1$.

$LHS = (1 + \tan^2\theta) \times \cos^2\theta = \sec^2\theta \times \cos^2\theta$ $[1 + \tan^2\theta = \sec^2\theta]$ (2)

$= (1/\cos^2\theta) \times \cos^2\theta = 1 = \text{RHS}$ (1)

Q. 4 Solve any three sub-questions. (9)

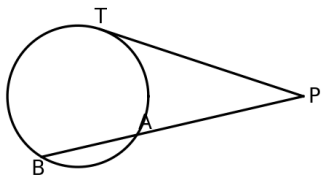
(i) In ΔABC, ray AD bisects ∠BAC and D lies on side BC. If AB = 10 cm, AC = 14 cm and BC = 6 cm, find BD and DC.

AD bisects ∠A, so BD/DC = AB/AC = 10/14 = 5/7 (Angle bisector theorem) (1)

Let BD = 5k, DC = 7k. Then 5k + 7k = 6 ⇒ k = 1/2 (1)

Ans: BD = 2.5 cm, DC = 3.5 cm (1)

(ii) In the figure, PT is a tangent to the circle at T and PAB is a secant. If PT = 12 cm and PA = 8 cm, find AB.



Tangent–secant theorem: $PT^2 = PA \times PB$ (1)

144 = 8 × PB ⇒ PB = 18 cm (1)

Ans: AB = PB − PA = 18 − 8 = 10 cm (1)

(iii) Using slopes, show that the points A(−2, −3), B(1, 0) and C(4, 3) are collinear.

Slope of AB = (0 − (−3))/(1 − (−2)) = 3/3 = 1 (1)

Slope of BC = (3 − 0)/(4 − 1) = 3/3 = 1 (1)

Slope of AB = slope of BC and B is common, so A, B, C are collinear. (1)

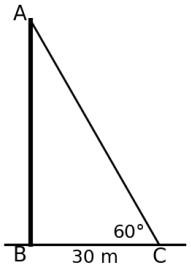
(iv) The radius of a circle is 6 cm. Find the length of the arc and the area of the sector corresponding to a central angle of 60°. (π = 3.14)

Length of arc = (θ/360) × 2πr = (60/360) × 2 × 3.14 × 6 = 6.28 cm (1½)

Area of sector = (θ/360) × πr² = (60/360) × 3.14 × 36 = 18.84 cm² (1½)

Q. 5 Solve any two sub-questions. (6)

(i) In the figure, AB is a vertical tower. From a point C on the ground, 30 m away from the foot B of the tower, the angle of elevation of the top A is 60°. Find the height of the tower. (√3 = 1.73)



In right-angled ΔABC, tan 60° = AB/BC (1)

√3 = AB/30 ⇒ AB = 30√3 = 30 × 1.73 (1)

Ans: Height of tower = 51.9 m (1)

(ii) The radius of a cone is 6 cm and its height is 8 cm. Find (a) its slant height, (b) its curved surface area, (c) its total surface area. (π = 3.14)

(a) $l = \sqrt{(r^2 + h^2)} = \sqrt{(36 + 64)} = 10$ cm (1)

(b) Curved surface area = πrl = 3.14 × 6 × 10 = 188.4 cm² (1)

(c) Total surface area = πr(r + l) = 3.14 × 6 × 16 = 301.44 cm² (1)

(iii) In ΔPQR, ∠Q = 90° and QS ⊥ PR. If PS = 4 cm and SR = 9 cm, find QS and PQ.

Theorem of geometric mean: $QS^2 = PS \times SR = 4 \times 9 = 36 \Rightarrow QS = 6$ cm (1½)

PR = PS + SR = 13 cm. $PQ^2 = PS \times PR = 4 \times 13 = 52$ (1)

Ans: PQ = √52 = 2√13 cm (½)

— End of Answer Key —